

Kunal Keshan

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POSITIONING SUMMARY

Software engineer with a full-stack, infrastructure-leaning skill set: comfortable across React/Next.js frontends, Node.js/NestJS backends, CMS-driven content platforms, and GCP/AWS cloud infrastructure (IAM, CDNs, load balancers, Docker, CI/CD). Architected and built the NestJS backend for a live, production ride-hailing platform (Zion Taxi) end-to-end, including payments (Razorpay), RBAC/security, a fare-pricing engine, and a Jest-driven unit test suite (231 backend + 57 frontend spec files), alongside its React admin dashboard. Experience spans a fast-moving product/infra role at a multi-vertical startup, fintech contract work building AI-assisted tools, and a run of independent freelance builds across healthcare, e-commerce, hospitality, and personal branding. Comfortable owning a project end-to-end, from infra and backend to frontend and deployment, and equally comfortable going deep on one layer within a larger team. Open to full-stack, backend-leaning, or platform/infra-focused roles, at startups or larger organizations.

SKILLS BANK

Languages: JavaScript (ES6+), TypeScript, Python, SQL, Bash/Shell, MATLAB, HTML5, CSS3

Frontend/Mobile: React.js, Next.js (App Router, SSG, ISR, on-demand revalidation), React Native, Expo, Module Federation (micro-frontend architecture), Tailwind CSS, shadcn/ui, Radix UI, Material UI, GSAP, Framer Motion / Motion, TanStack Query, Zustand, Redux, React Hook Form, Zod, Embla Carousel

Backend: Node.js, Express.js, NestJS, Python (FastAPI), Prisma ORM, Drizzle ORM, REST APIs, API architecture & design, Socket.io (realtime/WebSockets), FFmpeg (server-side video processing), ONNX Runtime (ML inference deployment)

Testing: Jest, ts-jest, Supertest, unit testing, test-driven development on service/business logic

Payments & Security: Razorpay (payment gateway integration), RBAC (role-based access control, permission grants, audit-actor coverage), WebAuthn, PII masking/data protection, CIDR-based access control

CMS/Content: Sanity, Payload CMS, Directus, Wix CMS API, Portable Text, React Email

Cloud & Infrastructure: Google Cloud Platform (IAM policies, load balancers, CDN, Compute Engine, Artifact Registry, Cloud Storage, secret management), AWS (S3, EC2, CloudFront), Cloudflare, Docker & Docker Compose (multi-stage builds, container networking, submodule-based builds), Nginx (reverse proxy, virtual hosts, TLS/ACME via Certbot), Vercel, Supabase, Grafana (agent-based monitoring)

Auth & Security: Auth0, Better Auth (multi-tenant RBAC via organization/admin plugins), Google Sign-In, Apple Sign-In, OAuth, Google reCAPTCHA v3

AI/ML: OpenAI API, IBM watsonx Orchestrate, LangChain, LangGraph (stateful agent orchestration), Retrieval-Augmented Generation (RAG) with pgvector (hybrid semantic + full-text retrieval), Langfuse (LLM observability/tracing, incl. self-hosted), adaptive filtering / signal processing (MATLAB), agentic assistant design

Messaging/Integrations: WhatsApp Cloud API (webhooks, interactive messages, Flow encryption), Telegram Bot API

DevOps/Tooling: Bash scripting & automation, GitHub Actions (CI/CD), Git submodules, build tagging & versioned deploys, Postman, Google Analytics (custom event tracking)

Databases: MongoDB, MySQL, PostgreSQL (via Prisma/Drizzle/Supabase), Redis

Collaboration & Communication: Cross-functional coordination (engineering and operations), developer mentorship, technical documentation (code-level, onboarding, business-facing), stakeholder communication, operations support

Other: Ngrok, web scraping, Nodemailer, i18n (next-intl), fuse.js (fuzzy string matching)

EXPERIENCE

StejasSYS

Software Engineer (Full-time)

Jan 2025 – Present

Chennai, Tamil Nadu, India

- Architected and built the NestJS backend for **Zion Taxi**, StejasSYS's in-house ride-hailing product, live in production with 300+ drivers, processing 100 rides/month (outstation and one-way-drop trips). Owned module boundaries, the fare/pricing-strategy pattern, caching/coherency design, and the RBAC permission model end-to-end.
- Implemented the platform's transactional core: Razorpay payment gateway integration, commission/ledger settlement (GST-exclusive), and rider debt/balance resolution.
- Built a security-hardened access layer: RBAC with granular permission grants and audit-actor coverage, PII-masking interceptors, WebAuthn device-carry, and CIDR-based access restriction.

- Wrote and maintain the platform's unit test suite using Jest: 231 backend spec files (payments, RBAC/access control, fare pricing, PII masking, realtime WebSocket gateways) and 57 frontend test files on the companion React admin dashboard (permissions, fare display, booking/audit logic); 3,700 individual test cases passing.
- Built the React 19/Next.js admin dashboard (**taxi-web-admin**) for internal ops: live trip tracking, fare/commission management, and driver/rider administration.
- Built the React Native/Expo driver and rider apps for Zion Taxi (the native client surface for the platform's 300+ drivers), including live GPS trip tracking and custom heading-calculation logic for smooth map navigation (device-heading smoothing for the driver app, route-direction-based fallback for the rider app), plus shared fare-breakdown logic reused across both apps.
- Built the conversation orchestration core, RAG-based knowledge retrieval system (pgvector), and admin API for **Innkraft's AI hotel sales agent**, a LangChain/LangGraph-powered WhatsApp/Telegram assistant handling guest conversations and booking flows for 5 hotel clients (more onboarding). Built the Simplotel booking-provider integration (one of three live provider integrations) and a 60+ script regression-eval framework serving as the system's primary conversation-quality safety net.
- Own the company's cloud infrastructure end-to-end: GCP IAM policies, load balancers, CDNs, and secret management across the company's core product lines, including ride-hailing/taxi, hospitality (Innkraft, The Grand Regent), restaurant (Bytenosh, Culinex), and wellness/travel/Plugginn (an embeddable widget platform built on Module Federation for the embedding logic; handled deployment there and stayed closely involved in its architecture and planning).
- Built the 4-language (English, Tamil, Hindi, German) website for **The Grand Regent** hotel client using next-intl locale routing, migrating its CMS backend from Sanity to Payload as the platform evolved.
- Full-stack ownership of **Culinex**, an in-house restaurant operations platform (multi-outlet inventory, waste tracking, production-batch tracking, POS sales-data reconciliation) currently under active development, spanning frontend components, application routes, and the database schema.
- Designed and implemented the platform's RBAC foundation (Better Auth organization plugin with custom permission statements) for multi-tenant access control.
- Joined the platform's order-recommendation system (a cron-based restock-quantity model using consumption/waste/guest-count data) already in progress, and contributed to its planning and feature development going forward.
- Designed and documented an intelligent sales-data mapping system for reconciling CSV exports from external POS systems via fuzzy item matching, reusable across outlets and POS types.
- Led infrastructure and backend setup for multiple projects, including deploying and maintaining Directus CMS instances with build tagging, code freezing, and version control across environments.
- Migrated the shared reverse-proxy stack from a single monolithic Nginx config to a per-domain `conf.d/` structure (confirmed via `sys-proxy/nginx.conf.old`, 471 lines, vs. the current split config), improving maintainability across the company's multiple hosted domains.
- Built `update-service.sh`, a Bash automation that pulls a new versioned image tag from GCP Artifact Registry, updates the Docker Compose service definition, and restarts the affected containers, replacing a manual multi-step deploy with one command; used across the proxy repos (`sys-proxy`, `bytenosh-proxy`, etc.).
- Set up GitHub Actions CI/CD (confirmed live in `wellness-web` and `plugginn`) alongside Bash-based local dev/deploy tooling.
- Deployed and operate a computer-vision ML inference service (fingertip-video-based vitals estimation, built by a teammate) via containerized FastAPI, ONNX Runtime, GCS video storage, and an Nginx reverse-proxy layer.
- Authored technical documentation across three audiences: code-level references, developer onboarding guides, and business/feature-facing docs explaining shipped functionality in non-technical terms, reducing ambiguity for new engineers and non-technical stakeholders alike.
- Drafted a company-wide engineering standards handbook (Definition of Done, code review/testing/security practices, and role-based onboarding guides), scaffolded as a documentation site; a personal initiative, not yet formally adopted org-wide.
- Coordinated day-to-day with a 3-5 person cross-functional engineering team and the on-site operations team running Zion Taxi's driver/rider operations, including 2-3 months directly fielding operations support requests as the platform's engineer during the product's early rollout.
- Mentor new developers and drive continuous improvement in architecture, process, and team productivity while staying hands-on with engineering work.
- **Stack:** NestJS, Jest, Razorpay, RBAC/WebAuthn, Redis, Socket.io, Drizzle ORM, PostgreSQL, React 19, Next.js, GCP, Docker/Docker Compose, Nginx, Directus, GitHub Actions, Bash, LangChain, LangGraph, pgvector, WhatsApp Cloud API, Telegram Bot API, Langfuse, Better Auth, Prisma, fuse.js (fuzzy matching), React Native, Expo, react-native-maps, expo-location, Module Federation, FastAPI, ONNX Runtime

Software Developer (Part-time)

Jul 2024 – Dec 2024

- Built full-stack features using Next.js, TypeScript, Tailwind CSS, and Payload CMS, leading the transition from low-code tooling (Wix) to custom-built solutions.
- Implemented SSG with on-demand ISR and CMS webhook integrations for seamless content sync between Payload and the frontend.
- Diagnosed and fixed a major performance bottleneck: optimized image handling and disabled unnecessary prefetching, cutting page load time from **15s to 1-2s**.
- Reduced Docker image bloat via multi-stage builds, meaningfully improving build efficiency and deployment speed.
- Gained hands-on experience with GCP services (Compute Engine, Artifact Registry, Cloud Storage), Docker networking, and multi-environment pipelines with internal API routing for cost optimization.
- **Stack:** Next.js, TypeScript, Tailwind CSS, Payload CMS, Docker, GCP

Flookup Capital Advisors (One21.ai)

Sep 2023 – Present

Contract Software Engineer

Mumbai, Maharashtra, India (remote)

- Built and deployed custom websites and web apps for finance & tech clients, primarily Chartered Accountant-related services, under the One21.ai product umbrella.
- Led the transition from no-code platforms (e.g., Webflow \$14/month) to custom Next.js deployments on Vercel, meaningfully cutting recurring hosting costs.
- Delivered dynamic CMS-driven solutions using Sanity, Payload, and Directus, letting non-technical client teams update content independently.
- Developed and scaled backend solutions on AWS, Supabase, and Vercel, including load balancers, S3 storage, and CloudFront CDN for optimized asset delivery.
- Implemented custom Google Analytics event tracking to capture business-specific user interactions and insights.
- Shipped multiple production solutions including: **MCA Tech** (mcatech.one21.ai), a business intelligence platform with real-time financial data, risk assessments, and analytics across a database of 2.6M+ companies; **CAGPT** (cagpt.in), an AI-powered platform delivering finance, taxation, compliance, and business-law guidance for accounting professionals; **Snapshot** (snapshot.one21.ai), a business intelligence platform tracking funding trends and company data across 200+ companies; **Merchant Banker** (merchantbanker.in); **Flookup** (flookup.com); **Valuation Work** (valuation.work); and **Captable, Due Diligence, Value Umbrella, GDS, Jobs**, additional One21.ai product suite tools.
- **Stack:** Next.js, TypeScript, Sanity, Payload, Directus, AWS, Supabase, Vercel

LingoScriptAI

Jul 2023 – Mar 2024

Product Development Engineer

Chennai, Tamil Nadu, India

- Led development on a content-creation tool, working the full Agile cycle from research and development through deployment.
- Built the web application using TypeScript, Next.js, Tailwind CSS, Auth0, Node.js, and Express.js.
- Implemented FFMPEG for core functionality including server-side video rendering.
- Gained hands-on experience with AWS S3 and EC2, contributing to the tool's successful launch; collaborated closely with the team to resolve blockers quickly.
- No formal usage metrics captured for this role: primarily a fast-paced learning and shipping period.
- **Stack:** TypeScript, Next.js, Tailwind CSS, Auth0, Node.js, Express.js, FFMPEG, AWS

Sundar Clinic

Apr 2023 – Sep 2023

Full-stack Developer

Chennai, Tamil Nadu, India

- Built the clinic's responsive website from scratch, establishing its first proper online presence.
- Integrated Sanity CMS so clinic staff could update site content independently, without developer involvement.
- Introduced Tamil and Hindi localization (via next-intl) alongside the existing English site, extending reach to a wider local audience.
- Conducted performance optimization for faster page loads on sundarclinic.com.
- Contributed to a **30% increase in clinic visits**, alongside a concurrent paid-ads push (the language localization work is believed to be a contributing factor, not the sole driver).
- **Stack:** Next.js, Sanity CMS, next-intl

SRM University

Nov 2022 – Mar 2023

Project Assistant

Chennai, Tamil Nadu, India

- Built a custom user interface (React, TypeScript, Axios) for a neural network model running on PYNQ Z-Series hardware, improving accessibility and interaction with the model.
- Collaborated with hardware engineers and medical professionals to integrate the UI successfully with the underlying hardware.
- Implemented secure Ngrok tunneling to reliably route traffic despite hardware limitations.
- **Stack:** React, TypeScript, Axios, Ngrok

Blackwins Tech Solutions

May 2022 – Jul 2022

Frontend Developer (Intern)

Chennai, Tamil Nadu, India

- Built user-facing interfaces for a SaaS product using React.js and Material UI, improving user engagement and satisfaction.
- Collaborated with designers and product managers to translate requirements into shipped features.
- Optimized performance to reduce server bandwidth usage during concurrent logins, improving load times and scalability.
- Conducted testing and debugging to resolve issues before release.
- No specific metric captured for this internship.
- **Stack:** React.js, Material UI

CLUB & COMMUNITY LEADERSHIP

Grouped separately from professional experience above: these were unpaid, community/student-org leadership roles run in parallel with coursework.

IEEE SRMIST

Oct 2021 – Sep 2023 (2 years)

- **Chairperson, Computer Society** (Feb 2023 – Sep 2023): Managed and mentored 50+ developers and AI/ML learners; collaborated with industry experts to bring advanced AI/ML techniques into real-world projects; led several internal skill-building projects.
- **Technical Lead, Web and App Domain** (May 2022 – Feb 2023): Mentored 25+ developers; led the technical side of 4+ events, including building event websites and handling registrations.
- **Vice Lead, Web Domain** (Sep 2021 – Feb 2022): Led workshops on HTML, CSS, JS, Node.js, Express.js, Git/GitHub, and MySQL; mentored 10+ developers; contributed to an internal club platform serving 200+ members.
- **Web Developer** (Jan 2021 – Sep 2021, IEEE Webtute track): Learned web development fundamentals; contributed to club projects; built a mini full-stack social media app (Node.js, Express.js, MySQL).

Think-Digital, SRM

Sep 2021 – Apr 2023 (2 years 4 months)

- **Team Head** (Oct 2022 – Apr 2023): Mentored and managed 40+ developers on full-stack fundamentals and advanced concepts; ran weekly cross-team meetings with other domain leads; partnered with Think-Digital to bring in more internship opportunities for the team.
- **Web Domain Lead** (Feb 2022 – Oct 2022): Mentored 10 developers; delivered "Atom," an in-house project used by 200+ community members; ran online workshops on frontend, backend, and Git/GitHub.
- **Vice Lead, Web Domain** (Sep 2021 – Feb 2022): a distinct, concurrent role from the same-titled IEEE SRMIST position above (confirmed the two ran in parallel on genuinely overlapping timelines, not a data error).

SRMpedia

Jan 2022 – Jul 2022

Web Developer

- Revived a broken mobile-app web scraper, restoring access to campus information for 3,000+ SRM students in KTR.

CodeChef SRM Student Chapter

Dec 2021 – Jul 2022

Web Developer

PROJECTS / FREELANCE

Ordered newest to oldest by primary build/activity date.

9wealth: Personal Finance / Wealth Management App (unlaunched) | React Native, Expo, NestJS, Prisma, Supabase,

- Building (not yet launched) a personal finance app for tracking budget, investments, assets, and net worth, with an AI chat interface for spending/investment insights; built across React Native/Expo (mobile), NestJS/Prisma/Supabase (backend), and OpenAI API (AI trend-chat), with infrastructure owned end-to-end.

Kumar Srivathsan: Portfolio Site | Next.js 16, TypeScript, Tailwind CSS, shadcn/ui, Vercel

Mar 2026

- Built a personal portfolio site for a maritime Navigation Officer, showcasing his career across container ships, Ro-Ro vessels, and international shipping routes. Live at kumarsrivathsan.com.

Community Knowledge Weaver: Agentic AI Assistant | *IBM watsonx Orchestrate, MCP, TypeScript* Jan 2026

- Hackathon project, IBM Dev Day "AI Demystified" Hackathon 2026. Built an agentic AI assistant unifying scattered organizational knowledge (projects, tickets, policies, notes) into one chat interface on IBM watsonx Orchestrate, with a multi-agent orchestrator routing each question to the right specialist agent, able to both answer and take action (create tickets, add notes) via MCP tools.

Keeping It Sou: Artist Website | *Next.js 16, TypeScript, Tailwind CSS, Sanity CMS, Vercel* Jan 2026

- Built the official website for hip-hop artist Sou, showcasing releases, music, and upcoming drops, with Sanity CMS for content management. Live at keepingitsou.com.

Dr. Nidhi: Portfolio Site | *Next.js, TypeScript, Tailwind CSS, GSAP, Framer Motion* Dec 2025

- Built a portfolio website for a doctor's practice, using GSAP and Framer Motion for polished page interactions.

Hzel Brown: Dessert Brand E-commerce (unlaunched) | *Next.js 16, React 19, TypeScript, Sanity CMS, Tailwind CSS, R*

- Built (not yet launched) an e-commerce platform for an artisan dessert brand (handcrafted brownies, brookies, cupcakes, and cookies with delivery across Tamil Nadu), with full menu/cart/categorization, Sanity CMS webhook-driven revalidation, and React Email transactional templates.

Junglithenomad: Wix CMS Video Carousel | *Next.js 14, TanStack Query, Wix CMS/Data API, Embla Carousel* Jul 2024 -

- Built a custom, server-hydrated video testimonial carousel pulling content natively from 2,500+ videos in the client's existing Wix CMS, embedded into their site via iframe, with lazy loading and infinite pagination for performance across the large catalog.

Additional smaller/older self-directed and academic projects, kept as a compact list:

Mind-Check (2023-2026): Web app assessing emotional well-being via the Burns Depression Checklist.

Shiryoku (2022-2025): Open-source curated resource list for upskilling across programming, design, and engineering.

Hardware Web UI (2022-2024): Web-based GUI for interfacing with a breast cancer detection neural network.

Xilinx Homepage (2023-2024): Site for SRMIST's Xilinx-funded Women in Technology (WIT) initiative.

EDUCATION

SRM Institute of Science and Technology

Bachelor of Technology (B.Tech), Electronics & Communication Engineering

- CGPA: 8.65

Sep 2020 – May 2025

Chennai, India

St. John's International Residential School

Jul 2013 – Apr 2020

CERTIFICATIONS & COURSES

SQL for Data Science, Coursera / UC Davis, Apr 2023

Covers core SQL for data work: SELECT/filtering/sorting (WHERE, BETWEEN, IN, LIKE, ORDER BY, GROUP BY), subqueries, joins and unions, and data-manipulation functions for analysis (the foundational course in UC Davis's "Learn SQL Basics for Data Science" specialization).

Responsive Web Design, freeCodeCamp, Aug 2021

Project-based certification (15 practice projects + 5 certification projects, 300 hours) covering semantic HTML, forms, accessibility, CSS Flexbox, and CSS Grid for fully responsive layouts.

M001: MongoDB Basics, MongoDB University, Jan 2022

6-chapter course covering MongoDB fundamentals vs. relational DBs, BSON/JSON, importing/exporting and querying data, document CRUD operations, indexing and the aggregation pipeline, and the Atlas/Compass tooling ecosystem.

Claude 101, Anthropic Academy, Aug 2026 (exact day pending)

Anthropic's foundational course on using Claude for everyday work tasks and understanding its core features and capabilities.

Claude Code 101, Anthropic Academy, Aug 2026 (exact day pending)

Anthropic's course on using Claude Code effectively within a daily development workflow.

Postman API Fundamentals Student Expert, Postman, Dec 2022

Introduction to Systems Engineering, Coursera, Nov 2022

Programming, Data Structures and Algorithms using Python, NPTEL, Sep 2022

The Node.js Master Class, Pirple, Jul 2022

Keeping Up With the JavaScripts: ES6, Pirple, Jan 2022

Web Development, Internshala, Oct 2021

AWARDS / RECOGNITION

Publication: "Leaky LMS Algorithm Based Low Complexity Adaptive Noise Cancellation," 2025 International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence (RAEEUCCI), Chennai, India, Apr 2025. DOI: 10.1109/RAEEUCCI63961.2025.11048209. Authors: S. Maiti, D. Adusumalli, K. Keshan, S. Sharma, S. H. Pauline, S. Dhanalakshmi. Proposes a multistage Leaky LMS adaptive filtering approach for low-complexity noise cancellation in biomedical (PCG) and speech signals, applied to denoising speech recordings from Parkinson's-affected patients to improve diagnostic signal quality. Implemented in MATLAB (repo: [Leaky-LMS-Algorithm-Based-Low-Complexity-Adaptive-Noise-Cancellation](#), built on earlier work in [PCG-Signal-Analysis-With-Adaptive-Filtering](#), Aug 2024, confirmed as the precursor project).

EXTRAS

Languages: English (Full Professional), Tamil (Native/Bilingual), Hindi (Native/Bilingual), Japanese (Elementary)